(iii) Measure of skewness based on Moments

The Measure of skewness based on moments is denoted by β_1 and is given by

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Example 6.18

Find β_1 for the following data $\mu_1 = 0$, $\mu_2 = 8.76$, $\mu_3 = -2.91$

Solution:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$\beta_1 = \frac{(-2.91)^2}{(8.76)^3} = \frac{8.47}{672.24}$$

$$= 0.0126$$

6.7.2 Kurtosis

Kurtosis in Greek means 'bulginess'. In statistics kurtosis refers to the degree of flatness or peakedness in the region about the mode of a frequency curve. The degree of kurtosis of distribution is measured relative to the peakedness of normal curve. In other words, measures of kurtosis tell us the extent of which a distribution is more peaked or flat-topped than the normal curve.

The following diagram illustrates the shape of three different curves mentioned below:

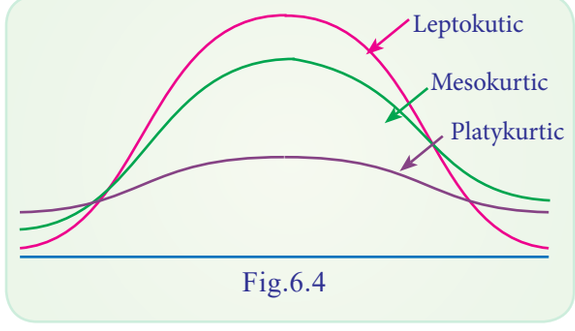


Fig.6.4

If a curve is more peaked than the normal curve, it is called 'leptokurtic'. In such a case items are more closely bunched around the mode. On the other hand if a curve is more flat-topped than the normal curve, it is called 'platykurtic'. The bell shaped normal curve itself is known as 'mesokurtic'. We can find how much the frequency curve is flatter than the normal curve using measure of kurtosis.

Measures of Kurtosis

The most important measure of kurtosis is the value of the coefficient. It is defined as: coefficient of kurtosis $\beta_2 = \frac{\mu_4}{\mu_2^2}$

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